

Analysis of the dynamic motion of ionic fluids (ionized water) on the basis of the theory of magnetohydrodynamics

To what extent does varying the temperature of an ionic fluid affect turbulence in the fluid, quantified by Reynolds Number and Hartmann's Number

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INTRODUCTION

This research paper is focused on answering the research question - **To what extent does varying the temperature of an ionic fluid affect turbulence in the fluid, quantified by Reynolds Number and Hartmann's Number.** By understanding fluid dynamics and propulsion of an ionic fluid this research question seeks to uncover the effect of temperature on the suppression of turbulence in ionized fluid while experiencing a force. Magnetohydrodynamics (MHD) a field of physics discovered by Nobel Prize winning physicist Hannes Aflén in 1970 is focused on the interaction between a magnetic field and a conducting fluid¹. The force (Lorentz Force) produced on a positive test charge due to the interaction between magnetic and electric fields² propels the fluid. MHD is purposed to dampen the uncontrollable (turbulent) flow of a fluid as laminar flow is more suitable for the application of mechanical physics and is energy conservative in nature which implies that not much energy is lost from the fluid³. The principle behind turbulence begins with an uncontrolled flow of a fluid that leads to the formation of eddies that dissipate energy by creating smaller eddies, which in turn create smaller eddies till all this energy is dissipated as thermal energy⁴. This energy loss amounts to a lot of potential energy converting to thermal unusable energy. Although in 1970 the discovery of magnetohydrodynamics resolved this issue by expanding on existing theories of physics. This research area has much potential to be applied in real life scenarios as it can make transportation in fluids, such as oceans and air, much more efficient and less resistant by the unpredictable state of these transportation mediums. More importantly plasma and liquid metals can be controlled by the same concept and therefore the application of MHD is vital in building fusion reactors in order to confine the turbulent eddies groomed in the plasma as it fuses. The independent variable being varying temperature was chosen in order to measure the proportionality between the increase in standard increments of this variable and notice if the turbulence damping caused by MHD can lead to the restoration of a steady or laminar flow or distort the existing laminar flow.

Initially my ideas were on the tangent of checking the effect of changing the geometry of the magnetic field that drives the MHD but upon extensive research i found that the strength of magnetic field changes with geometry at

¹ Magnetohydrodynamics.” *ScienceDirect*, Elsevier, <https://www.sciencedirect.com/topics/materials-science/magnetohydrodynamics>

² This interaction is known as cross fields

³ Müller, Ulrich, and Lars Bühler. *Magnetohydrodynamics in Channels and Containers*. Springer, 2001.

⁴ Eddy Motion.” *ScienceDirect*, Elsevier, <https://www.sciencedirect.com/topics/engineering/eddy-motion>.

strengths less than 1T and it is extremely difficult to replicate a magnetic strength of more than 1T in a school lab and also by making an electromagnet with high current therefore I researched further and discovered that a substantial effect on the thrust can be achieved by changing the temperature of the ionic fluid⁵. More research indicated that change in temperature affects the turbulence in a fluid as well which linked me back to my original research area⁶. My research question has not been formally investigated yet and hence there is no secondary data or literature that I have evaluated and have solely relied on primary data collected.

Ideally an increase in the temperature increases the strength of the Lorentz force that acts on the fluid and increases its velocity. The reason this makes for an interesting research area is because increase in temperature affects both Hartmann's number and Reynolds number and would also impact multiple variables in the formula due to which we will explore modeling methods and assumption in order to approximate the effect of temperature on these two dimensionless quantities and which has a greater effect on the fluid we are studying.

BACKGROUND PHYSICS

Defining Key Terms

Table 1.0 entails the definitions of all the key terminology used throughout this research paper with appropriate citation in order to refer for deeper understanding.

Key term	Definition	Citation
Magnetohydrodynamics	The study of the behavior of electrically conducting fluids , such as plasma, liquid metals and salt water, in the presence of magnetic fields. In MHD the motion of the fluid is affected by both electromagnetic forces and hydrodynamic forces .	Davidson, P. A. (2016). <i>An Introduction to Magnetohydrodynamics</i> (2nd ed.). Cambridge University Press.
Temperature	Temperature is the measure of the average kinetic energy of the	Halliday, D., Resnick,

⁵ Davidson, P. A. *An Introduction to Magnetohydrodynamics*. Cambridge University Press, 2001.

⁶ Qayyum, Sajid, et al. "Magnetohydrodynamic Analysis and Fast Calculation for Fractional Maxwell Fluid with Adjusted Dynamic Viscosity." *Applied Sciences*, vol. 10, no. 1, 2020, pp. 1–15, <https://doi.org/10.3390/app10010005>.

	particles of a fluid. It is a property that reflects how hot or cold a body is.	R., & Walker, J. (2014). Fundamentals of Physics (10th ed.). Wiley.
Eddy currents	Loops of electrical current induced within a conductor when it experiences a changing magnetic field. According to Faraday's law of electromagnetic induction, these circulating currents generate their own magnetic field which opposes the original change in the magnetic field. ⁷	Lorrain, P., Corson, D. R., & Lorrain, F. (2000). Electromagnetic Fields and Waves (3rd ed.). W.H. Freeman
Turbulence	Refers to a type of fluid flow characterized by chaotic and irregular motion where the velocity of the fluid at a given point varies over time. ⁸	<i>Pope, S. B. (2000). Turbulent Flows. Cambridge University Press.</i>
Lorentz force	The Lorentz force is a force exerted on charged particles moving through an electrically conductive medium in the presence of varying magnetic fields.	Griffiths, D. J. (2017). Introduction to Electrodynamics (4th ed.). Cambridge University Press.
Inertial force	Inertial force arises from the motion of the fluid and is related to the fluid's momentum. It represents the force due to which the fluid upholds its tendency to remain at rest or continue moving due to its	https://www.sciencedirect.com/topics/engineering/inertial-force

⁷ This concept led to the need to **use Hartmann Number** as increase in turbulence leads to greater eddy currents that generate their own eddy currents that may at some point overpower the applied magnetic field, damping the turbulent effect of MHD.

⁸ It is caused by high Reynolds numbers where **inertial forces dominate over viscous forces leading to complex vortices and eddies.**

	gained momentum. ⁹	
Viscous force	Refers to the internal friction with a fluid, arising from interaction between different layers of the fluid moving at different velocities . It acts to resist the relative motion between layers of the fluid leading to energy dissipation. ¹⁰	White, F. M. (2016). Fluid Mechanics (8th ed.). McGraw-Hill Education.
Electrodynamic forces (crossed fields)	They arise from the interaction between electric charges and magnetic fields. These forces are responsible for the motion of charged particles and can be described by Lorentz force law	Jackson, J. D. (1998). Classical Electrodynamics (3rd ed.). Wiley.

We already learned in the introduction what magnetohydrodynamics is. We will now evaluate the forces and the reason why the fluid is propelled in an MHD setup. The basics of MHD can be uncovered by understanding how an electric motor works.

Lorentz force in an electric motor¹¹

Lorentz force is a result of the interaction between magnetic field and electric current in a conductor. On a microscopic scale a charge is seen to experience a force in the presence of varying magnetic flux. This force can be defined as follows:

⁹ Inertial forces are the cause of increase in turbulence. When **Inertial force** >> **viscous force** flow is more turbulent.

¹⁰ Viscous forces oppose inertial forces to dampen turbulence and restore laminar flow.

¹¹ **National High Magnetic Field Laboratory.** "Lorentz Force." *Magnet Academy*, National MagLab, <https://nationalmaglab.org/education/magnet-academy>

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (1)$$

where:

- $\mathbf{F}$ is the Lorentz force,
- q is the electric charge of the particle,
- $\mathbf{E}$ is the electric field,
- $\mathbf{v}$ is the velocity of the charged particle,
- $\mathbf{B}$ is the magnetic flux density.

In an electrical motor a current carrying conductor cuts through the external magnetic field the conductor is placed in due to which the resulting force propels the wire, generating mechanical motion. The direction of the force is given by Flemmings left hand rule¹² as depicted in **Figure 1.0**

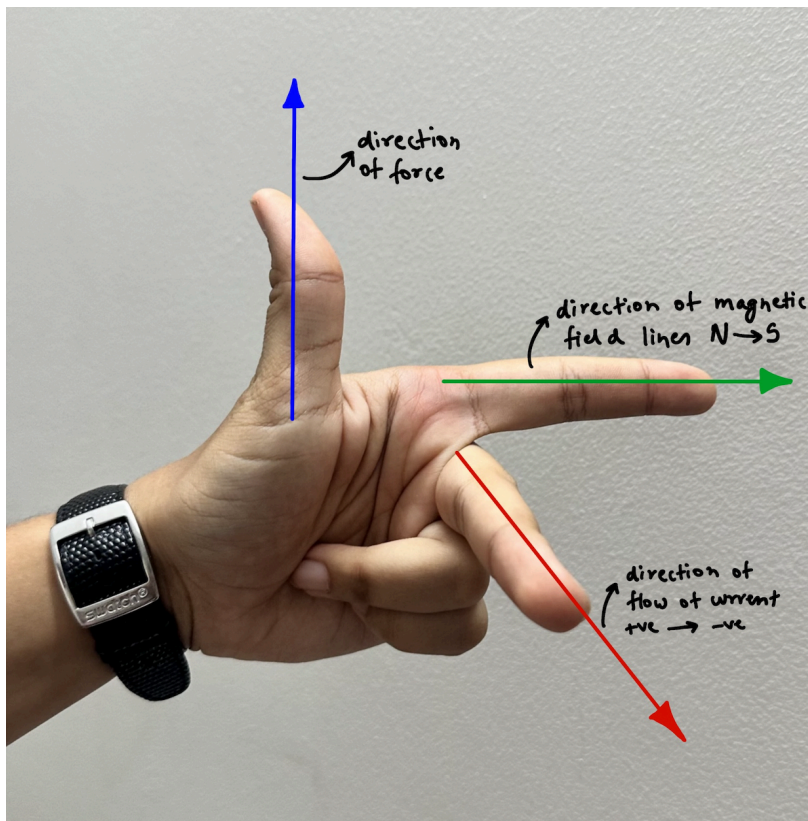


Figure 1.0 depiction of Fleming's Left hand rule (thumb middle and index finger mutually perpendicular to each other) depicting what each finger represents. When the index finger is aligned with the direction of magnetic field (north to south), middle finger is aligned with direction of flow of current (positive to negative electrode) the thumb gives the direction of the magic force experienced by a small positive charge.

¹² Pearson. Physics for the IB Diploma Program. 2023, Pearson Education Limited

This force called the Lorentz force generates a torque in the motor, converting electrical energy into mechanical work. The same principle is transferable to the experiment conducted by applying it to a conducting fluid in a magnetohydrodynamic system, but the medium of interaction changes from a solid wire to a fluid that can carry current.

Lorentz force and MHD

In an MHD system, instead of a solid wire, the conductor is an **ionic fluid**, such as an ionized liquid or plasma, that can carry electric current. The basic principles of the Lorentz force still apply, but the force now acts on the fluid itself. When a conducting fluid is introduced to a magnetic field, and a current is passed through the fluid (using appropriate electrodes) at the point where the magnetic field meets the fluid such that the direction of current is perpendicular to the line of magnetic field, the current interacts with the magnetic field, producing a force in accordance with Flemmings left hand rule.¹³

This interaction between current and magnetic field either enhances or suppresses the movement of the fluid, depending on the field strength and the properties of the fluid, such as temperature and electrical conductivity as depicted in **Figure 1.1**. The effects of these forces are directly related to turbulence in the fluid, which is the focus of this study.

¹³ This theory provides justification for the setup of the experiment and the choice of NaCl solution as the ionic liquid.

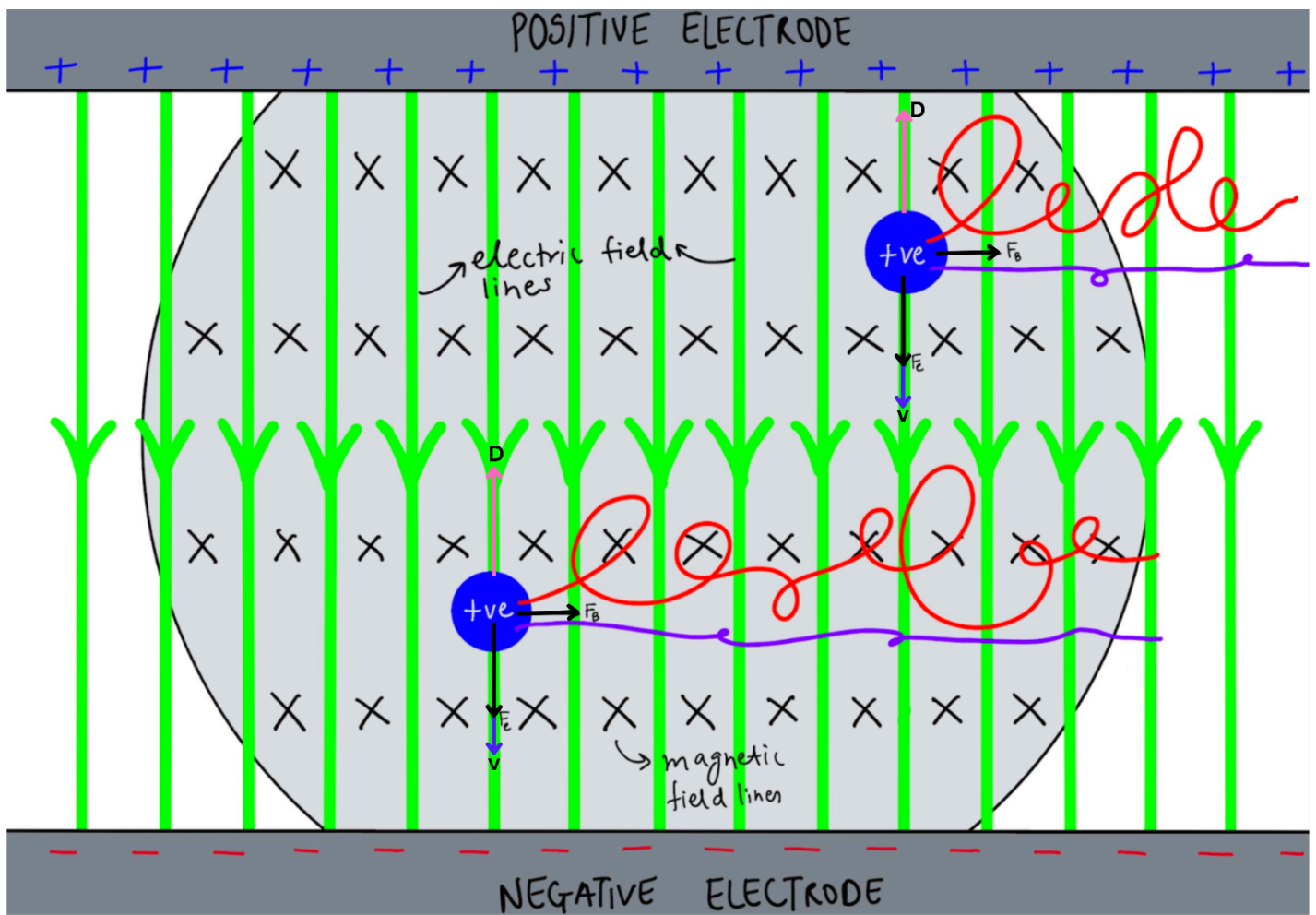


Fig 1.1 depicts the interaction between the magnetic field (depicted by the X suggesting the field is going into the plane of the screen) and electric fields (green arrowed lines) perpendicular to each other. This results in two forces acting on a small positive charge. The red line depicts the possible turbulent route that the field might force the particle to move in and the purple line depicts the possible laminar dampened route the field might force the particle to move in. The arrow V represents the initial motion of the particle and D represents the drag force.

Fundamental Equations and Parameters:

In order to determine the relationship between inertial forces and viscous forces within the ionic fluid at different temperatures Reynolds Number was chosen which is a unitless quantity proportional to the ratio between inertial forces and viscous forces. This ratio is critical in deducing the relationship between temperature and turbulence

as this constant gives us a scale to measure the turbulence in a fluid. Greater the temperature, greater is the average kinetic energy of particles in the fluid, greater is the velocity achieved by these particles, greater are the inertial forces that in turn leads to the formation of eddy currents that causes an increase in turbulence.

Reynolds Number (Re)¹⁴: Measures the ratio of inertial forces to viscous forces within a fluid. It is defined as:

$$Re = \frac{\rho u L}{\mu} \text{ — (2)}$$

- Where:
 - ρ = Fluid density (constant)
 - u = Flow velocity (dependent)
 - L = Characteristic length (constant)
 - μ = Dynamic viscosity (varies with temperature)
- The reason we are also interested in the Hartmann constant is because changing the temperature not only affects viscosity but also affects the fluid's conductivity and since MHD is dependent on the conductivity of the fluid we have to take this factor into account while understanding its effect on the characteristics of flow of the fluid. Moreover the increase in eddy currents that generate their own magnetic field would ideally increase the damping effect on the fluid's motion.

Hartmann Number (Ha)¹⁵: Represents the influence of a magnetic field on the fluid's motion. It is defined as the ratio between induced electrodynamic (magnetic) force and the hydrodynamic force of viscosity It is given by:

$$Ha = BL\sqrt{\frac{\sigma}{\mu}} \text{ — (3)}$$

¹⁴ Reynolds Number and Its Importance in Fluid Dynamics." Physics LibreTexts, 20 Oct. 2023, https://phys.libretexts.org/Bookshelves/Thermodynamics_and_Fluid_Mechanics/Fluid_Mechanics_Basics/Reynolds_Number.

2300 > Re indicates laminar flow , 2300 < Re < 4000 indicates the transition phase from laminar to turbulent and 4000 < Re indicates turbulent flow.

¹⁵ Nivedita, K. "The Role of Hartmann Number in Magnetohydrodynamics." ResearchGate, https://www.researchgate.net/publication/Role_of_Hartmann_Number_in_MHD

10 > Ha indicates weak influence of magnetic field on flow (turbulence governed by inertial and viscous force), 10 < Ha < 100 transition phase from turbulent to laminar (magnetic force begins to become significant), Ha > 100 indicates laminar flow as magnetic field strongly dampens turbulence.

- Where:
 - B = Magnetic field strength (constant)
 - L = Characteristic length (constant)
 - σ = electrical conductivity of the fluid
 - μ = Dynamic viscosity (varies with temperature)
- The Hartmann number quantifies the magnetic field's impact on fluid flow, with higher values indicating greater suppression of turbulence.

Influence of Temperature on Reynolds Number (Re) and Hartmann Number (Ha) : Temperature variation impact both the Reynolds number and the Hartmann number in the following ways:

- **Reynolds Number (Re):**
 - **Viscosity (μ):** Temperature is inversely proportional to viscosity. In the NaCl solution, the decrease in viscosity due to increase in temperature results in a higher Reynolds number, as lower viscosity means less resistance to flow, potentially leading to more turbulent behavior.
 - **Density (ρ):** Temperature changes also affect fluid density, but this impact is usually smaller compared to viscosity changes.

NOTE: The effect of temperature-induced density changes on the Reynolds number will be considered negligible compared to the primary effect of viscosity changes.

- **Hartmann Number (Ha):**
 - **Electrical Conductivity (σ):** Conductivity is directly proportional to temperature. An increase in conductivity enhances the Hartmann number, indicating a more significant effect of the magnetic field in suppressing turbulence.
 - **Dynamic Viscosity (μ):** While temperature reduces viscosity, the impact on the Hartmann number is secondary to changes in conductivity. The decreasing viscosity increases the Hartmann number, but its effect is less pronounced than that of conductivity.

Experimental Assumptions and Postulates:

For the ionized water in this experiment, the impact of temperature on density will be considered negligible compared to the effects of viscosity changes. The primary focus will be on how viscosity affects Reynolds number and how conductivity influences Hartmann number. Moreover temperature-induced changes in electrical conductivity are expected to play a more significant role in altering the Hartmann number than changes in viscosity. This assumption allows for a more straightforward analysis of how the magnetic field's effectiveness varies with temperature. Keeping the magnetic field strength constant throughout the experiment will also isolate its effects of temperature on the Reynolds and Hartmann numbers ensuring that any observed changes in flow behavior are due to temperature variations alone. Furthermore it is assumed that the temperature distribution within the ionized water is uniform during the experiment and will be conducted under steady-state conditions, where the flow of current in the fluid and magnetic field are stable over time. These assumptions ensure that the observed effects are due to temperature changes rather than transient phenomena and simplifies the analysis by avoiding complications arising from temperature gradients.

HYPOTHESIS

According to the background physics and the crux of this exploration we can consider the following changes observed in Reynolds number and the Hartmann number due to a change in temperature.

Hypothesis 1

As the temperature of the ionic fluid increases the Reynolds number will also increase. This because higher temperatures reduce the fluid's viscosity allowing it to flow more easily which in turn increases the velocity as the fluid resists to change in velocity less. This reduced resistance to flow will cause the inertial forces to dominate over viscous forces resulting in higher turbulence.

As the temperature of the ionic fluid increase the Reynolds Number will increase indication more turbulent flow

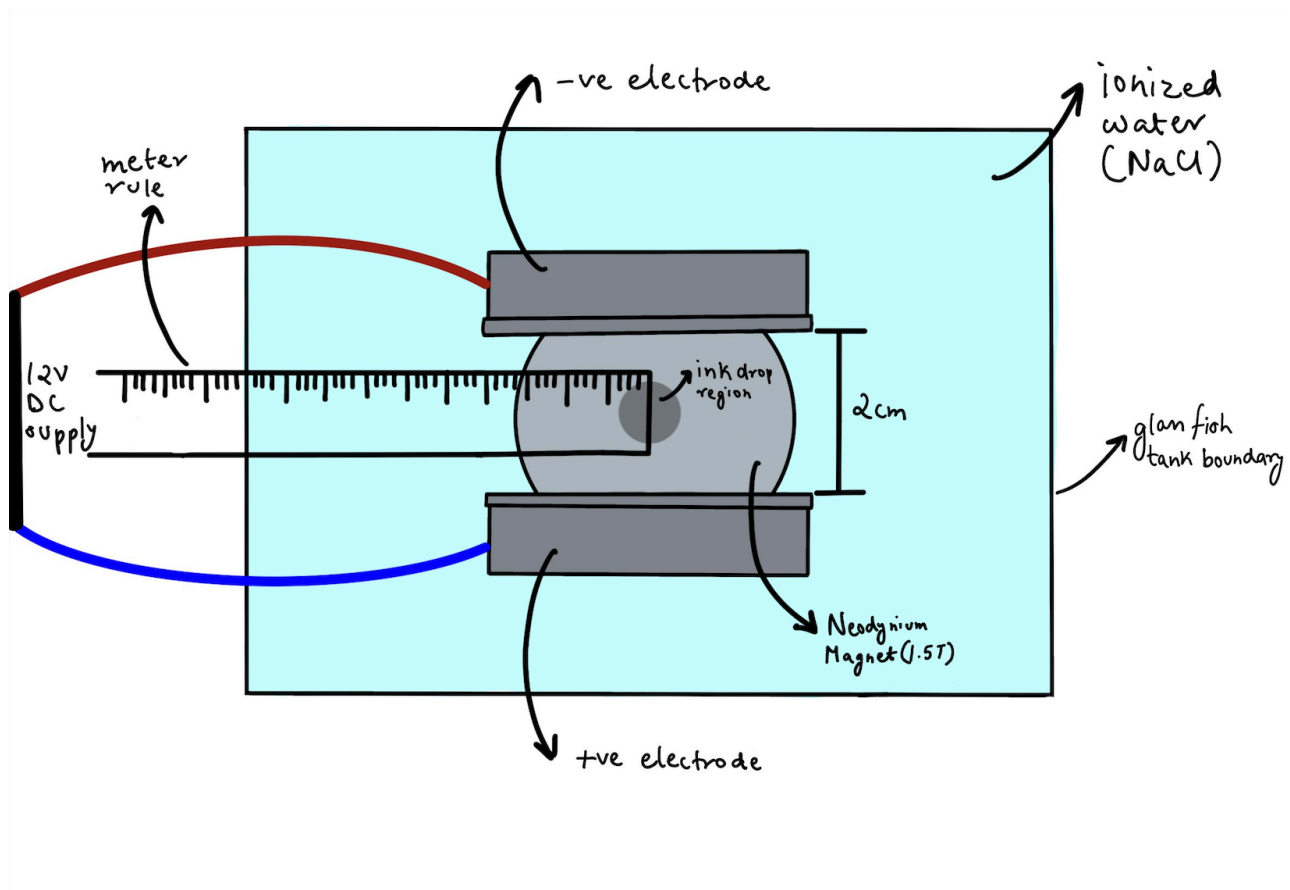
Hypothesis 2

As the temperature of the ionic fluid increases the Hartmann number will increase as the electrical conductivity of the fluid tends to increase with temperature which will enhance the interaction between the fluid and the constant magnetic field. This will therefore result in the magnetic field exerting a larger damping effect on the fluid potentially reducing the turbulence despite higher Reynolds Number.

As the temperature of the ionic fluid increases the Hartmann number will increase leading to stronger magnetic damping effects on the flow

METHODOLOGY AND PROCEDURE

Figure 1.2 depicts the aerial view of the set up used to conduct the magnetohydrodynamic propulsion experiment post the heating of the ionized fluid. The magnet and the ruler were both placed under the transparent tank.



Procedure¹⁶:

A ***transparent fish tank*** was taken and thoroughly cleaned before starting the experiment. ***Two tin plates*** were taken and bent at 90 degree forming an L shape as shown in **Figure 1.2** . These electrodes were secured centrally to the bottom of the tank at a fixed distance of 2 cm apart ensuring that the distance between the electrodes *L* remains constant¹⁷. *It was ensured that the electrodes are not completely submerged in the NaCl solution. A significantly **strong neodymium magnet** was placed at the bottom of the tank directly beneath the electrodes ensuring that the magnetic field was perpendicular to the electric field as shown in **Figure 1.1**. It was ensured that the orientation of the magnetic field (magnets pole in contact with the tank) does not change during the trials.* Next ***3 liters of water*** was measured using an appropriate ***measuring cylinder*** and added to the glass tank. Then ***50 grams of salt*** was measured and added to the water and stirred until fully dissolved ensuring that the salt is evenly distributed (homogeneous ionic fluid). The experiment began at room temperature (24.5°C). For increasing the independent variable by constant increments of 10°C a ***water bath*** was used and the temperature of the water was measured using a ***digital thermometer*** to ensure accuracy after heating. Post this the electrodes were connected to a ***12V DC desk power eliminator***. *It was necessary to ensure that the direction of current was kept constant. For each trial it was necessary to ensure that the **ink** was injected in about the same place and from the same height.* A viscometer probe and conductivity probe were calibrated by a logger pro and suspended into the tank ensuring they do not touch the bottom of the tank.

¹⁶ All the apparatus used has been made bold and italic and controlled variables have been made italic only

¹⁷ Link back to **Equation 3** in background information **pg 10**

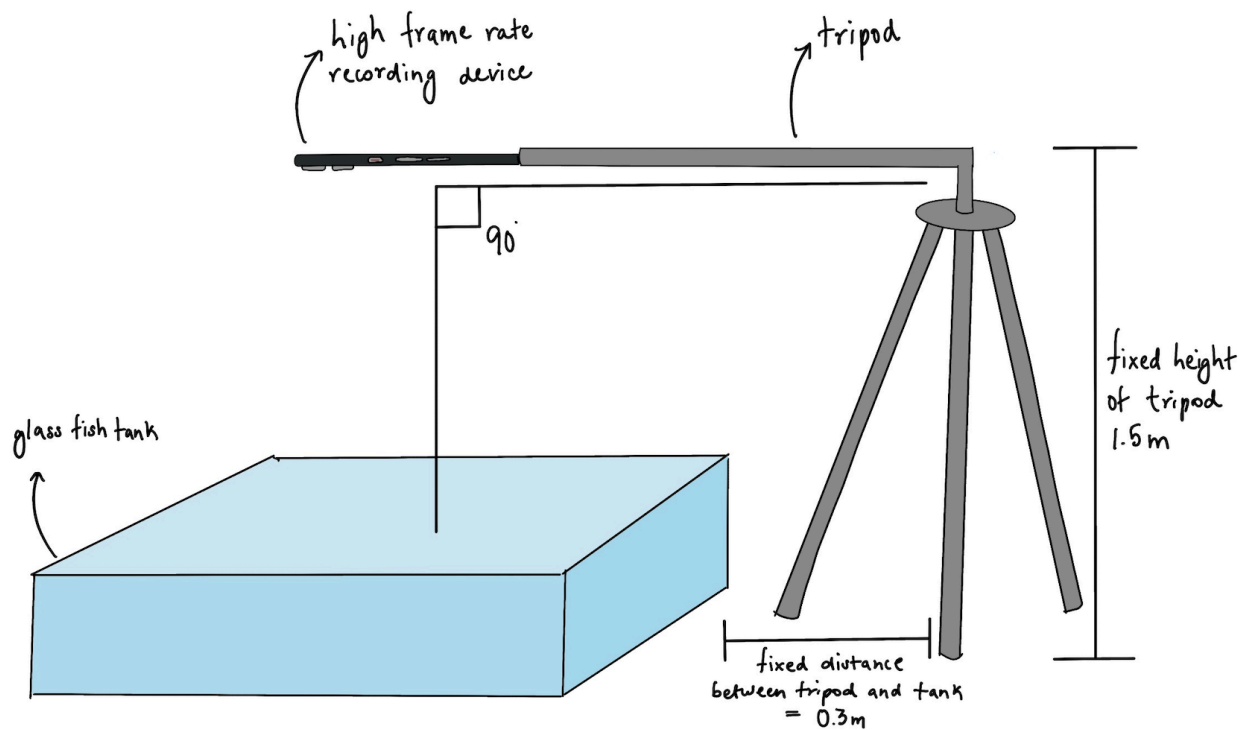


Figure 1.3 is a depiction of the placement of the tripod and the camera relative to the tank that the experiment was carried out in.

A **high frame rate camera** was used to record the progression of the experiment. The camera on the **tripod** was aligned perpendicular to the tank as shown in **Figure 1.3** to capture a clear view of the inks movements. *The distance and angle at which the camera is placed was kept constant throughout all trials in order to ensure accuracy and reproducibility.* The camera was then turned on and the eliminator was switched on applying a constant current of 12V across the two electrodes. *It was ensured that the value of current was kept constant throughout all trials.*

After recording the ink flow at each temperature the videos were uploaded to **vernier video analysis software** and the frame by frame analysis tool was used to track the distance covered by the ink in a given standard time that was later used to calculate the velocity of the fluid. The time frame of the analysis was taken to be 20 seconds after the eliminator was turned on to ensure that all the forces had balanced out and close to terminal velocity

could be measured. For each temperature it was ensured that the entire process was repeated three times to minimize the effect of random errors accounting for outliers.

RAW DATA TABLE

Table 1.1 holds all the raw data collected while conducting the experiment for 3 trials and 13 independent variables. (highest increment the water bath could go up to from initial room temperature). All values have been rounded off to 3 significant figures.¹⁸

	Trial 1				Trial 2				Trial 3			
SR NO	Temperature (T)/ °C ±0.05	Velocity (u)/ ms ⁻¹	Viscosity (μ) mPas	Conductivity (σ) S/cm	Temperature (T)/ °C ±0.05	Velocity (u)/ ms ⁻¹	Viscosity (μ) mPas	Conductivity (σ) S/cm	Temperature (T)/ °C ±0.05	Velocity (u)/ ms ⁻¹	Viscosity (μ) mPas	Conductivity (σ) S/cm
1	24.70	0.077	0.917	0.078	24.70	0.076	0.916	0.079	24.70	0.076	0.914	0.079
2	35.00	0.096	0.823	0.081	35.00	0.097	0.823	0.079	35.00	0.095	0.822	0.080
3	45.00	0.125	0.761	0.079	45.00	0.129	0.759	0.077	45.00	0.128	0.760	0.076
4	55.00	0.165	0.704	0.083	55.00	0.166	0.707	0.084	55.00	0.166	0.707	0.080
5	65.00	0.185	0.665	0.081	65.00	0.182	0.663	0.081	65.00	0.183	0.664	0.080
6	75.00	0.201	0.620	0.125	75.00	0.199	0.621	0.126	75.00	0.201	0.620	0.125
7	85.00	0.221	0.598	0.224	85.00	0.222	0.598	0.223	85.00	0.220	0.587	0.222
8	95.00	0.177	0.574	0.325	95.00	0.176	0.574	0.322	95.00	0.174	0.576	0.322
9	105.00	0.138	0.552	0.333	105.00	0.137	0.555	0.334	105.00	0.137	0.552	0.332

- ¹⁸¹. Viscosities of Common Fluids - Oil Viscosity Chart." *Oil Viscosity Chart*, <https://oilviscositychart.com/learn/viscosity-list.php>.
2. Helmenstine, Anne Marie. "Table of Electrical Resistivity and Conductivity." *ThoughtCo*, 12 May 2024, <https://www.thoughtco.com/table-of-electrical-resistivity-conductivity-608499>.
3. Seawater Properties." *Engineering Toolbox*, https://www.engineeringtoolbox.com/sea-water-properties-d_840.html.
4. Thermophysical Properties of Seawater." *Journal of Physical and Chemical Reference Data*, vol. 7, no. 3, 1978, pp. 941-948. National Institute of Standards and Technology (NIST), <https://srdata.nist.gov/JPCRD/jpcrd176.pdf>.

The above sites have been accessed to obtain literature values that have been used to find the average viscosity and conductivity values

10	115.00	0.112	0.541	0.376	115.00	0.112	0.540	0.377	115.00	0.112	0.542	0.376
11	125.00	0.105	0.530	0.440	125.00	0.104	0.530	0.437	125.00	0.103	0.530	0.437
12	135.00	0.104	0.519	0.495	135.00	0.105	0.521	0.496	135.00	0.107	0.519	0.495
13	145.00	0.106	0.510	0.564	145.00	0.107	0.509	0.565	145.00	0.106	0.507	0.566
14	155.00	0.108	0.496	0.635	155.00	0.108	0.497	0.634	155.00	0.106	0.496	0.635

TRANSFORMED DATA TABLE

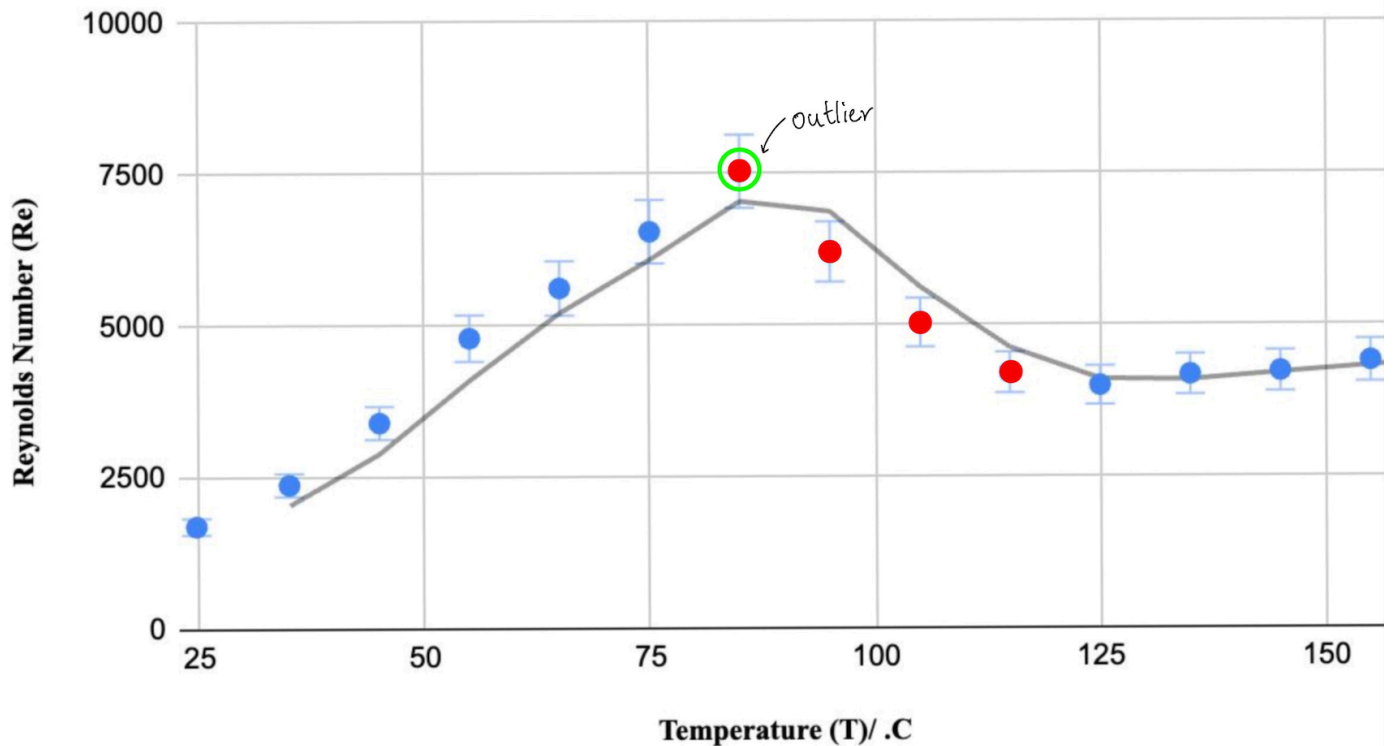
Table 1.3 entails the transformed data resulting in the values for Hartmann and Reynolds Number that have been used to graph graph 1.0 and 2.0 below. All values have been rounded off to 3 significant figures except Reynolds and Hartmann numbers.

SR NO.	Temperature (T)/ °C ±0. 05	Mean Velocity (u)/ ms ⁻¹ ±0. 0005	Mean Viscosity (μ) mPas ± 0. 0015	Mean Conductivity (σ) S/cm±0. 001	Reynolds Number (Re) ±0. 257	Hartmann Number (Ha) ±0. 276
1	24.70	0.0760	0.9150	0.079	1694.000	27.810
2	35.00	0.0960	0.8230	0.080	2376.000	30.630
3	45.00	0.1270	0.7600	0.077	3396.000	32.120
4	55.00	0.1660	0.707	0.082	4790.000	35.220
5	65.00	0.1830	0.6640	0.080	5612.000	39.930
6	75.00	0.2000	0.6210	0.125	6541.000	46.500
7	85.00	0.2210	0.5980	0.223	7532.000	54.920
8	95.00	0.1750	0.5750	0.323	6203.000	62.050
9	105.00	0.1370	0.5530	0.333	5039.000	73.620

10	115.00	0.1120	0.5410	0.376	4207.000	79.130
11	125.00	0.1040	0.5300	0.438	3994.000	86.200
12	135.00	0.1060	0.5190	0.495	4175.000	92.610
13	145.00	0.1060	0.5080	0.565	4230.000	100.040
14	155.00	0.1070	0.4960	0.635	4400.000	107.330

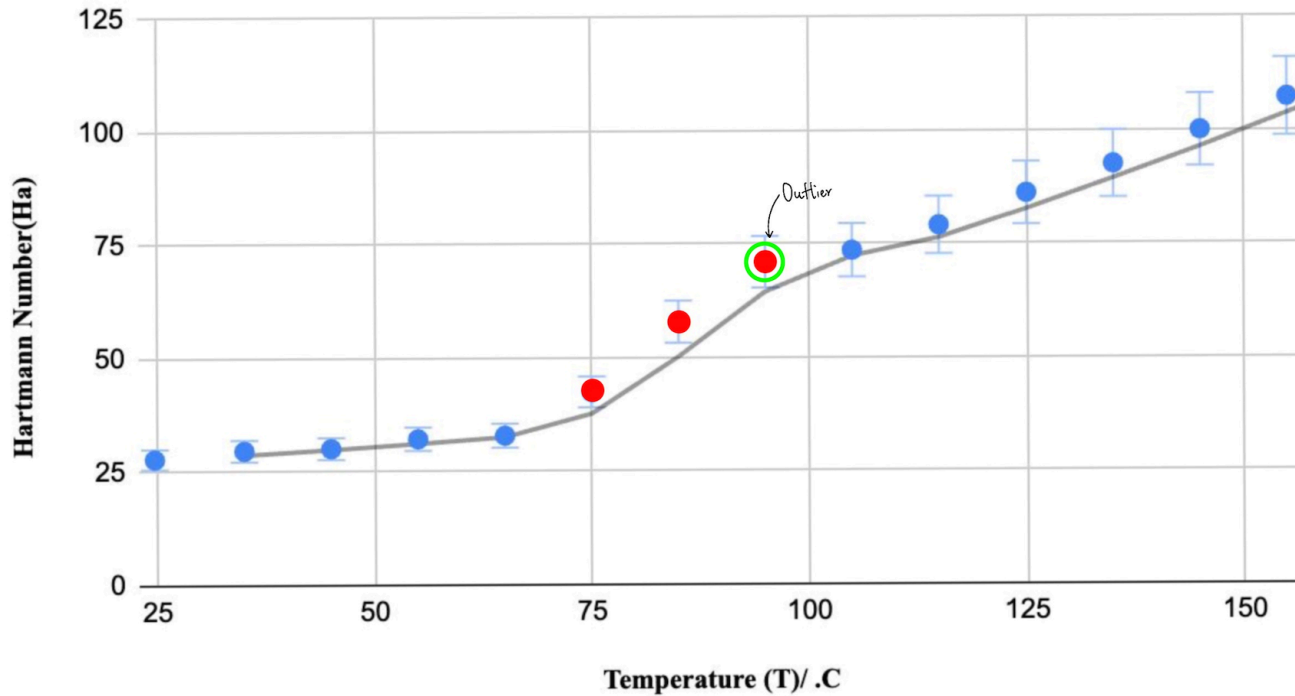
Graph 1.0 depicts the trend of Reynolds numbers plotted against temperature. Uncertainties are depicted as a percentage of the value justifying the varying size of error bars.

Reynolds Number (Re) vs. Temperature (T)/ .C



Graph 2.0 depicts the trend of Hartmann numbers plotted against temperature. Uncertainties are depicted as a percentage of the value justifying the varying size of error bars.

Hartmann Number(Ha) vs. Temperature (T)/ .C



UNCERTAINTY PROPAGATION

Table 1.4 - Sample calculation for mean viscosity conductivity and velocity, Reynolds and Hartmann number and its associated uncertainty for $T = 105\text{ }^{\circ}\text{C}$

Calculation	Uncertainty
1) finding average velocity and its uncertainty $u_1 = 0.138\text{ ms}^{-1}$ $u_2 = 0.137\text{ ms}^{-1}$ $u_3 = 0.137\text{ ms}^{-1}$	Uncertainty propagation for u_{mean} $\Delta u_{mean} = \frac{max-min}{2}$ $\Delta u_{mean} = \frac{0.138-0.137}{2}$ $\Delta u_{mean} = \pm 0.0005$

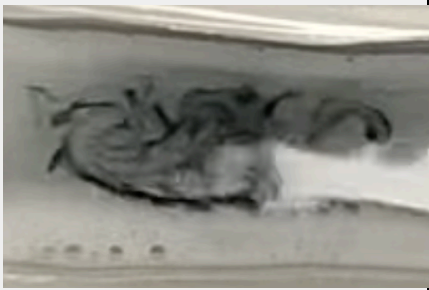
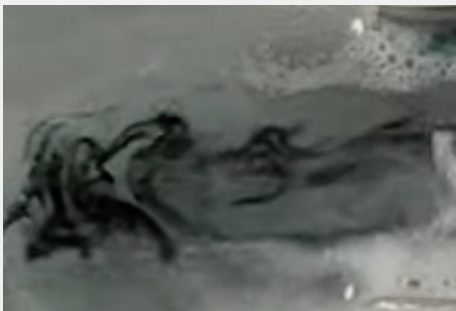
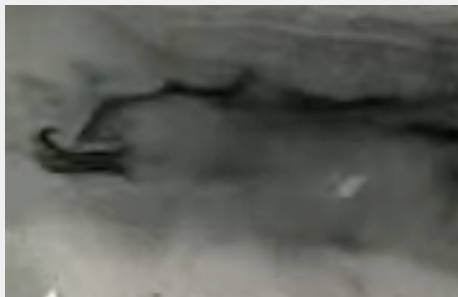
$u_{mean} = \frac{u_1 + u_2 + u_3}{3}$ $u_{mean} = \frac{0.138 + 0.137 + 0.137}{3}$ $u_{mean} = 0.137 \pm 0.0005 \text{ ms}^{-1}$	
2) finding average viscosity and its uncertainty $\mu_1 = 0.552 \text{ mPas}$ $\mu_2 = 0.555 \text{ mPas}$ $\mu_3 = 0.552 \text{ mPas}$ $\mu_{mean} = \frac{\mu_1 + \mu_2 + \mu_3}{3}$ $\mu_{mean} = \frac{0.552 + 0.555 + 0.552}{3}$ $\mu_{mean} = 0.553 \pm 0.0015 \text{ mPas}$	Uncertainty propagation for u_{mean} $\Delta\mu_{mean} = \frac{max-min}{2}$ $\Delta\mu_{mean} = \frac{0.555-0.552}{2}$ $\Delta\mu_{mean} = \pm 0.0015$
3) finding average conductivity and its uncertainty $\sigma_1 = 0.333 \text{ Scm}^{-1}$ $\sigma_2 = 0.334 \text{ Scm}^{-1}$ $\sigma_3 = 0.332 \text{ Scm}^{-1}$ $\sigma_{mean} = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$ $\sigma_{mean} = \frac{0.333 + 0.334 + 0.332}{3}$ $\sigma_{mean} = 0.333 \text{ Scm}^{-1}$	Uncertainty propagation for σ_{mean} $\Delta\sigma_{mean} = \frac{max-min}{2}$ $\Delta\sigma_{mean} = \frac{0.334-0.332}{2}$ $\Delta\sigma_{mean} = \pm 0.001$
4) finding average Reynolds Number value and its uncertainty $Re = \frac{\rho u L}{\mu}$	Uncertainty propagation for Re $\Delta Re = \left(\frac{\Delta\rho}{\rho} + \frac{\Delta u}{u} + \frac{\Delta L}{L} + \frac{\Delta\mu}{\mu} \right)$

$Re = \frac{1017 \times 0.137 \times 0.02}{0.553 \times 10^{-3}}$ $Re = 5039$	$\Delta Re = \left(\frac{0.53}{1017} + \frac{0.0005}{0.137} + \frac{0.005}{0.02} + \frac{0.0015}{0.552} \right)$ $\Delta Re = \pm 0.257$
5) finding average Hartmann Number and its uncertainty $Ha = BL\sqrt{\frac{\sigma}{\mu}}$ $Ha = 1.5 \times 2\sqrt{\frac{0.333}{0.553 \times 10^{-3}}}$ $Ha = 73.62$	Uncertainty propagation for Ha $\Delta Ha = \left(\frac{\Delta \sigma}{\sigma} + \frac{\Delta L}{L} + \frac{\Delta \mu}{\mu} + \frac{\Delta B}{B} \right)$ $\Delta Ha = \left(\frac{0.001}{0.333} + \frac{0.005}{0.02} + \frac{0.0015}{0.552} + \frac{0.03}{1.5} \right)$ $\Delta Ha = \pm 0.276$

QUALITATIVE DATA (FEW SAMPLES)

Table 1.5 entails images of a few independent variables (35 °C, 45 °C, 55 °C, 65 °C, 95 °C, 105 °C, 115 °C, 135 °C, 155 °C) values at t= 5 seconds after the ink was added to the fluid depicting the eddy currents formed that are qualitative proof depicting turbulence in the fluid depicting and increase in vortex density with increasing temperature but less spiraling and circular flow path for 115 °C, 135 °C, 155 °C.

Temperature °C	35	45	55
Image of turbulence vortices			
Temperature °C	65	95	105

Image of turbulence vortices			
Temperature °C	115	135	155
Image of turbulence vortices			

CONCLUSION

The experiment successfully demonstrates the complex relationship between temperature, turbulence, and magnetic damping in an ionized fluid, with a distinct shift occurring after 85°C, where magnetic damping effects began to significantly suppress turbulence. This was quantified by analyzing both the Reynolds and Hartmann numbers as the temperature of the fluid increased.

Reynolds Number: Increasing Turbulence with Temperature

In accordance with **Graph 1.0** it is evident that initially as the temperature of the ionized solution increased, the Reynolds number exhibited a progressive positive correlation. This rise in Reynolds number indicated that the fluid became more turbulent as its temperature increased. The rationale behind this observation lies in the fact that, at higher temperatures, the viscosity of the fluid decreases, allowing the fluid to flow more easily and experience less internal resistance and contributing to the increasing Reynolds number in this range, as the fluid's

resistance to flow diminishes, allowing for higher speeds and greater inertial forces.. This is consistent with the hypothesis, as lower viscosity at higher temperatures allows the inertial forces of the fluid to dominate over the viscous forces, leading to greater chaotic motion within the fluid.

Although, the graph only supports the predicted pattern up to around 85°C. The proportional increase in the fluid's turbulence is disrupted as it undergoes an unpredicted exponential decrease till 125°C depicted by the red data points in **Graph 1.0** thus depicting values that indicate a transition from turbulent to laminar flow. Moreover the rate of increase in Reynolds Number varies vastly before and after the rapid decrease in the value. The initial rate is much higher than the rate of increase after the drop. The visual depiction of decrease in eddies seen in **Table 1.5** where temperatures *105 °C, 115 °C, 135 °C, 155 °C* illustrates less “swirling motion” that characterizes eddy currents and turbulence as the temperature increases. The pictures show the ink flowing in a more uniform than circular path that along with all the other evidence suggests that one of the forces, either the magnetic or electric force, acting on the ions are constraining its motion from erratic and turbulent to linear and laminar.

Hartmann Number

While the Reynolds number reflects the increasing turbulence with temperature, the Hartmann number in accordance with **Graph 2.0** represents the influence of the magnetic field on the fluid, also steadily increasing proportionally with increase in temperature as the ions of NaCl salt dissociate better increasing the conductivity. While this is aligned with hypothesis 2 the slope of the initial rate of Hartmann's number is very low due to which its impact on stronger stabilizing effect of the magnetic field is insignificant and the inertial forces continue to dominate as depicted by the Reynolds Number trend in **Graph 1.0**. However, the critical transition occurs after 85°C, where the influence of the magnetic field, represented by the Hartmann number, begins to play a more significant role in controlling the fluid's behavior. At higher temperatures, the fluid's conductivity improves, resulting in the formation of more vortices that leads to increased eddy currents. These eddy currents have their own magnetic field amplifying the existing magnetic field's ability to dampen the chaotic motion caused by the increased turbulence.

From both **Graphs 1.0 and 2.0** it is evident that after the temperature crosses 85°C , a marked change occurs in the relationship between the Reynolds and Hartmann numbers. The previously rapid increase in the Reynolds number begins to slow down, indicating that the turbulence, while still present, is being controlled by the stronger magnetic field. The Hartmann number's rise signifies that the magnetic damping effect has become strong enough to overpower the inertial forces that cause the fluid to become more turbulent at higher temperatures.

Furthermore more qualitative proof of this is depicted through images of ink vortices formed during MHD propulsion cumulated in **Table 1.5**. As temperature is increased the ink is seen to follow more erratic and circular paths clustering closer but from temperature 95°C to 155°C the ink spreads out forming fewer circles and flowing linear and parallel to the electrodes (less erratically) depicting reduced turbulence and progression towards laminar flow.

This shift is crucial because it marks the point at which the magnetic damping effect becomes dominant. Although the fluid's temperature continues to rise, the magnetic field exerts a stabilizing force that reduces the chaotic motion, effectively counteracting the fluid's tendency toward greater turbulence. The amplified electromagnetic forces due to eddy currents can create local turbulence or structured flow patterns between subsequent layers of the fluid mimicking increased viscosity. The dampening of the vertical motion along the forces as depicted in **Figure 1.1** constrains the particles to move along a linear path decreasing the velocity of the particle opposing its inertial motion.

This outcome aligns with the underlying principles of magnetohydrodynamics (MHD), where magnetic fields can suppress turbulence and stabilize fluid motion, especially when the fluid is sufficiently conductive. In this case, the observed behavior of the Reynolds number confirms the expected initial rise in turbulence as temperature increases, but the key observation is the subsequent suppression of this turbulence due to the growing influence of the magnetic field and the increasing conductivity of the fluid at higher temperatures allowing the magnetic field to exert a stronger influence, ultimately leading to reduced turbulence as reflected by the Hartmann number.

EVALUATION

The use of colored ink to track fluid motion was a great advantage as this allowed for a more intuitive understanding of flow behavior, through visual patterns of turbulent vortices highlighting irregularities characterizing turbulent and laminar flow more naturally than a solid object would have. Furthermore a solid object could have introduced additional forces like drag or resistance that would have influenced the flow hindering analysis of the effect of temperature. However a major drawback of using ink was its rapid diffusion in water, especially at higher temperatures. As the temperature increased, the ink diffused too quickly, making it difficult to track in order to determine the velocity of the propulsion.

The method of increasing the temperature in 10°C increments (from 10°C to 155°C) was a good approach, as it provided clear snapshots of the fluid's behavior across a broad temperature range. Moreover, the data range studied was vast enough to evaluate and discover a critical point whose trend could not have been verified if the data was limited to 100°C . Although the transfer of water from the water bath to the experimental tank introduced a notable temperature loss, which affected the precision of the temperature control in the experiment. Since transferring the heated water took time, it cooled down before entering the tank, meaning that the actual temperature in the tank might not have been the intended one. This discrepancy likely contributed to inconsistencies in the data, especially at higher temperatures where small changes in temperature can have more pronounced effects on fluid behavior¹⁹.

Using a neodymium magnet instead of an electromagnet as intended earlier as part of the apparatus ensured a more consistent magnetic field, making the analysis of Hartmann number behavior more reliable. This helped isolate the effects of temperature on fluid turbulence and damping, providing better insight into the fluid's magnetohydrodynamic behavior. Moreover the Hartmann number and the Reynold number are both appropriate choices as indicators of turbulence in the chosen context and experiment as while they both compliment each other depicting the impact of temperature on their individual values they also helped identify and justify the

¹⁹ This was taken into consideration and was accounted in the assumptions section of the background information **pg 11**

critical point at which a shift in linearity was observed providing results that was expected through this experiment.

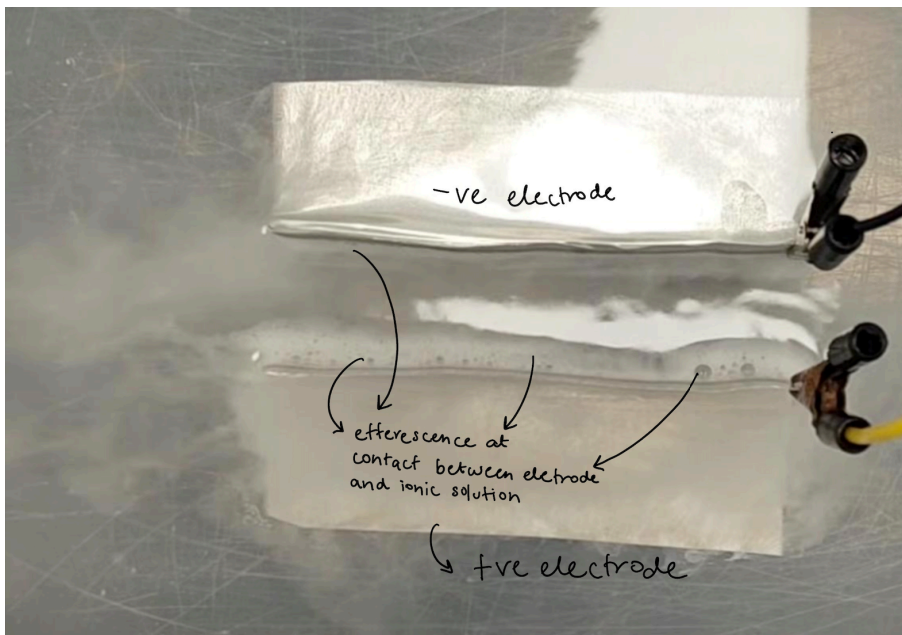


Figure 1.4 is an annotated diagram depicting effervescence / bubbling observed at the contact surface between electrodes.

Contrastingly the electrolysis of water during the experiment produced bubbles near the electrodes as depicted in **Figure 1.4** , which hindered the fluid's motion making it harder to track the true behavior of the fluid under the influence of the temperature change. These bubbles caused irregularities in the flow, which could have impacted the accuracy of the observed results. The presence of outliers in the data suggest that there were irregularities during certain trials which consistently grew at higher temperatures verifying the existence of the above errors affecting the flow in a way that wasn't accounted for by the available apparatus.

Final Implications for MHD Systems

The experiment was successful as it verified my hypothesis and also went beyond to give conclusive evidence of a way to decrease the turbulence of a fluid. This can be taken further in the real world by experimenting with the temperature of the object that needs to be propelled and whether that results in laminar flow as well or whether the temperature difference between the fluid and the propelled object is what affects its flow.

The fact that magnetic damping becomes dominant after 85°C has important implications for industrial and scientific applications of magnetohydrodynamics. In high-temperature environments, where turbulence can become a significant issue, the ability of magnetic fields to control and suppress turbulence offers a practical solution for managing fluid flow in MHD systems. I believe that my paper has an irrefutable relevance in real world fluid engineering where controlling turbulence is critical, such as in fusion reactors, plasma confinement, or liquid metal cooling systems in advanced technologies.

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